

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

Criteria	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						

Code of Conduct:

I G Madhulatha(192311295)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

G Madhulatha

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains "Offer" (Yes/No)
- Contains "Free" (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Q1. Identify TP, FP, FN, TN

Considering “Wolf present” as Positive:

- **True Positive (TP):** Wolf is present and villagers correctly respond → **0**
- **False Positive (FP):** Boy cries “Wolf” when no wolf is present, and villagers respond → **2**
- **False Negative (FN):** Wolf is actually present, but villagers do not respond → **1**
- **True Negative (TN):** No wolf is present and villagers correctly do not respond → **0**

Answer: TP = 0, FP = 2, FN = 1, TN = 0

Q2. Cat Image Classification

Given:

- Total cat images = **30**
- Model correctly predicts cats in 25 images → **TP = 25**
- Cat images missed = $30 - 25$ → **FN = 5**
- Total non-cat images = **70**
- Correctly predicts absence of cat in 50 images → **TN = 50**
- Incorrectly predicts cat in remaining images = $70 - 50$ → **FP = 20**

Confusion Matrix Values

TP = 25, FP = 20, FN = 5, TN = 50

Precision

Precision = 55.56%

Recall

Recall = 83.33%

F1-Score

F1-Score = 66.67%

Q3. Naïve Bayes Email Classification

1. Prior Probabilities

Total emails = **6**

- Spam = **3**

- Not Spam = 3

Answer:

- $P(\text{Spam}) = 0.5$
- $P(\text{Not Spam}) = 0.5$

2. Conditional Probabilities

For Spam

There are 3 Spam emails:

Feature	Probability
$P(\text{Offer}=\text{Yes} \mid \text{Spam})$	$2/3$
$P(\text{Offer}=\text{No} \mid \text{Spam})$	$1/3$
$P(\text{Free}=\text{Yes} \mid \text{Spam})$	$3/3 = 1$
$P(\text{Free}=\text{No} \mid \text{Spam})$	$0/3 = 0$
$P(\text{Short} \mid \text{Spam})$	$2/3$
$P(\text{Long} \mid \text{Spam})$	$1/3$

For Not Spam

There are 3 Not Spam emails:

Feature	Probability
$P(\text{Offer}=\text{Yes} \mid \text{Not Spam})$	$1/3$
$P(\text{Offer}=\text{No} \mid \text{Not Spam})$	$2/3$
$P(\text{Free}=\text{Yes} \mid \text{Not Spam})$	$0/3 = 0$
$P(\text{Free}=\text{No} \mid \text{Not Spam})$	$3/3 = 1$
$P(\text{Short} \mid \text{Not Spam})$	$1/3$
$P(\text{Long} \mid \text{Not Spam})$	$2/3$

3. Classify (Offer=Yes, Free=No, Length=Short)

For **Spam**:

For **Not Spam**:

Prediction: NOT SPAM

Because $0.0556 > 0$.

4. WEKA Comparison

Using the **Naïve Bayes classifier in WEKA**, the dataset can be loaded and the classifier run to compare its prediction. With the given data and without smoothing, the expected prediction is **Not Spam**, matching the manual calculation.

5. Independence Assumption

Naïve Bayes assumes that features are **independent of each other given the class**. For example, it assumes that **“Offer,” “Free,” and Email Length** independently affect whether an email is Spam or Not Spam.

In reality, these features may be related. For example, spam emails containing **“Offer”** may also often contain **“Free.”** This assumption may not always be completely true, but Naïve Bayes can still perform effectively for classification.